

Aanidhya Patidar

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SUMMARY

CS (AI & ML) student who builds end-to-end — fine-tuned PyTorch classifiers, deployed FastAPI backends, production PWAs with real users. Two shipped projects in agritech, one in deepfake detection. Looking for a backend or ML internship where the problems are hard and the systems actually run in production.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science (AI & ML)

GPA: 8.71/10.0

Bhopal, MP

2023 – 2027

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), SQL

Frameworks & Libraries: FastAPI, Flask, React, Vite, PyTorch, Pandas, NumPy, Scikit-learn, LightGBM, SQLAlchemy, TanStack Query, i18next, Recharts, Sentence-Transformers

Databases: PostgreSQL, SQLite, MySQL

Developer Tools: Git, GitHub, VS Code, Postman, Alembic, Docker, Vercel, Render, WebSockets, REST APIs

PROJECTS

Krishi Khata — Digital Farm Ledger & Market Intelligence PWA | *FastAPI, React, Vite, PostgreSQL, Gemini AI* | [live](#) | [github](#) 2025 – 2026

- Built and deployed a mobile-first PWA for Indian farmers — digital accounting ledgers, real-time mandi price tracking, and AI crop advisory. Frontend on Vercel, backend on Render, backed by Neon PostgreSQL.
- Designed a JIT mandi price pipeline using government APIs, caching 30-day commodity histories per district in PostgreSQL for sub-200ms lookups — replacing wasteful batch scraping entirely.
- Integrated a bilingual AI engine via Google Gemini — the backend intercepts `Accept-Language` headers and rewrites Gemini system instructions at runtime, streaming native Hindi or English responses without storing translated text.
- Set up a hardened WebSocket gateway for real-time community chat with JWT-authenticated handshakes, pre-upgrade rejection of unauthorized connections, and SlowAPI rate limiting on write-heavy routes.
- Localized the React frontend across 70+ UI strings via i18next, added searchable Combobox dropdowns for mandi queries, and built a Quick-Select Crop Hub that auto-fills search parameters from the farmer's active crops.

DeepGuard — Deepfake Face Detection System | *Python, PyTorch, EfficientNet-B4, OpenCV, MTCNN*

2025

- Fine-tuned EfficientNet-B4 on FaceForensics++ to classify real vs. AI-manipulated faces — dataset covers face-swap and reenactment forgeries across multiple compression levels.
- Built an MTCNN face extraction pipeline in OpenCV to crop and align face regions before classification, cutting background noise and tightening the model's attention to the actual manipulation zone.
- Applied Grad-CAM to surface which facial regions (mouth corners, eye edges, blending seams) drove predictions — made outputs interpretable beyond a raw confidence score, useful for debugging failure cases.
- Handled class imbalance via weighted cross-entropy loss; used stratified splits across forgery subtypes to stop the model from over-indexing on one manipulation method. Evaluated on held-out test set using AUC-ROC alongside accuracy.

AgroSense v4 — Crop Yield Prediction Platform | *Python, LightGBM, Flask, React, Open-Meteo API*

2025

- ML & Backend Lead on a 6-person team — sole owner of model training, feature engineering, and Flask REST API design; coordinated cross-functional deliverables across frontend and data sourcing sub-teams.
- Built a LightGBM regression model on historical crop, soil, and rainfall data across multiple crops and districts; evaluated using MAE and R^2 per crop category rather than in aggregate.

ACTIVITIES

Google Developer Groups on Campus — VIT Bhopal

2024 – Present

LeetCode: Solved 200+ problems across arrays, graphs, dynamic programming, and trees — leetcode.com/u/SG6rCPbiIR

CERTIFICATIONS

- Applied Machine Learning in Python** — University of Michigan (Coursera)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**